

Matt Bowring

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[Website](#) | [Github](#) | [LinkedIn](#)

Experience

The MathWorks

Software Engineer (Math/PDE)

MATLAB, C++, Python, CMake, OpenMP, Git

May 2022 - Present

Natick, MA (on-site)

- Lead development of the [MATLAB Quantum Computing Library](#) (unreleased prototype to 9,100+ users)
 - Designed object-oriented MATLAB framework to enable quantum programming across Windows, Linux, and Mac
 - Developed templated C++ state-vector simulator for CPU/GPU with OpenMP multi-threading
 - Implemented graph-based C++ compiler/transpiler to generate optimized gate-level assembly code
 - Built client-side REST infrastructure to manage async hardware jobs using AWS and IBM cloud services
 - Developed sparse algorithms for estimation of Pauli observables, error mitigation, and gate decomposition
 - Manage CI/CD, write all unit/functional/performance tests, and benchmark against other quantum libraries
- Consult industry users on machine learning and quantum computing applications
 - Implemented quantum neural networks for image classification ([SPIE publication](#)) and reinforcement learning
 - Researched constraint-aware quantum optimization methods based on Pauli correlation encoding, quadratization, and penalty dephasing
 - Developed presentations and examples for industry conferences (ACS 25, AWS re: Invent 25, IBM QDC 22-25)

The MathWorks

Software Engineer

MATLAB, C++, Python, Git

May 2021 - May 2022

Natick, MA (on-site)

- Trained recurrent graph network on QM7-X molecular dataset to predict equilibrium energy and classify optimal configurations; Developed format-agnostic data pipeline to encode atomic properties using OpenBabel
- Developed REST interface to cloud-based quantum annealing hardware; Benchmarked minor-embedding heuristics on constrained combinatorial problems; Wrote parallel simulations to compute qubit transition probabilities for truncated graph Hamiltonian across coupling strength and energy separation parameters

Education

Purdue University (3.8 GPA)

M.S. Electrical/Computer Engineering

Jan. 2025 - Present

West Lafayette, IN (remote)

- Fabricated mixed-signal PCB to solve combinatorial problems using phase dynamics of LC oscillators ([arXiv preprint](#));
 - Developed simulations of coupled oscillators in LTspice and modeled noisy phase dynamics with MATLAB
 - Designed schematics and layouts for two multi-layer boards in Altium Designer and soldered components
 - Developed real-time embedded C++ firmware using SPI and FreeRTOS to measure phases at 1MHz
 - Automated test instruments with MATLAB to control perturbations and analyze Fourier spectrum

The University of New Hampshire (3.9 GPA)

B.S. Mechanical Engineering

Aug. 2017 - May 2021

Durham, NH (on-site)

- Awarded over \$100,000 in scholarships for developing an autonomous quadcopter
- Led the Quadcopter Engineering Team; Volunteered as Makerspace Administrator, Academic Tutor and Mentor

Projects

Computer Systems

2024 - Present

- Operate custom multi-GPU Linux machine to accelerate research experiments; Familiar with dataflow architectures and SIMD parallelism; Broad interest in programming language semantics

Machine Learning

2020 - Present

PyTorch/CUDA, NetworkX, OpenCV

- Develop various research experiments: analyzed "edge-of-chaos" regions for randomly initialized Transformers; implemented non-reversible Langevin MCMC for discrete/relaxed optimization; Wrote packet parser to process financial data using Wireshark; Trained CNN/RNN to predict dash-camera misalignment; Built data pipeline and Bayesian network to predict trends for online video game.